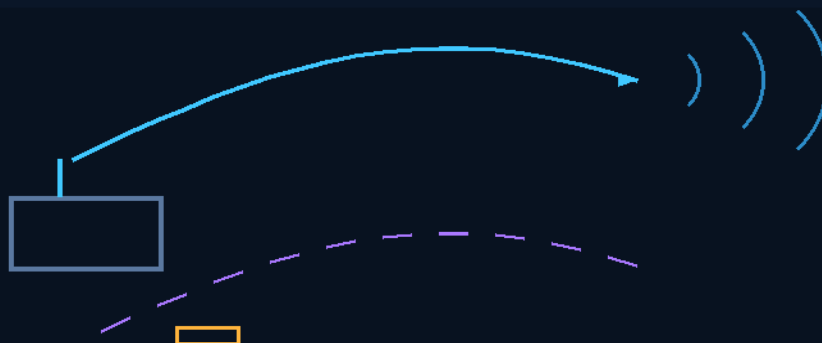


THE STARSHIP GALILEO
DEEP SPACE TRAVEL AND COLONIZATION
INSTRUCTIONS MANUAL
THE GEMINI PROTOCOLS

PHASE 15

BOW-RAIL REUSABLE MMWAVE RADAR "PHANTOM" NAVIGATION ROCKETS



FLIP 180° - BRAKE - REVERSE - CATCH

Fire, scan, flip, brake, catch, refuel, fire again.
The ship never flies blind.

GP-IM-015

v1.0 - 20 August 2026

STATUS: CURRENT

20 August 2026

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THE GEMINI PROTOCOLS — CHAPTER 15

PHASE 15: BOW-RAIL REUSABLE MMWAVE RADAR "PHANTOM" NAVIGATION ROCKETS — STATUS: CURRENT. The ship's forward eyes: reusable radar rockets fired from the bow rail on a 60-minute cadence, scanning thousands of miles ahead through dust and dark, feeding ultra-dense telemetry to the helm, then flipping 180 degrees, braking, and flying home into the underside hangar bay to be caught, refueled, and cycled forward again. Flanking mini-sats — the phantom sister ships — sweep the lateral void for blind-side hazards. Vetted: PASS — navigation concept, with the open items carried in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-015, v1.0 — 20 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the phantom law as seated (contents line and both design bullets, verbatim) — §2 why it works (the physics, graded with the vet's constraint) — §3 the system on the ship (the fire-and-catch cycle, the sister-ship divergence) — §4 the doctrine record (how the book already fights with these rockets)— §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE PHANTOM LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Launches short-range, reusable guidance rockets from the forward hull tracks. Rockets project high-density mmWave radar tracking lines through dense dust clouds or nebulas, charting safe navigation corridors ahead of the main ship.' [*VET CONSTRAINT, seated §5: read 'through dense dust clouds or nebulas' as mmWave forward reconnaissance in particulate environments — not a guarantee of penetrating every dense nebula; a nebula's density, ionisation and plasma properties vary enormously, and propagation is set by wavelength, particle size and concentration.*]

THE SEATED DESIGN BODY — FIRE-AND-CATCH (verbatim): 'Fire-and-Catch Navigation Probes: Reusable

mmWave radar rockets fired from the bow rail every 60 minutes. Probes scan thousands of miles ahead, transmit ultra-dense telemetry data back to the helm, flip 180°, brake via retrorockets, and reverse into an underside pass-through open vacuum hangar bay to be captured magnetically, refueled, and cycled back to the bow.'

THE SEATED DESIGN BODY — PHANTOM SISTER SHIPS (verbatim): 'Phantom Sister Ships: The vessel executes a 60-minute course divergence to Port and Starboard, ejecting radar mini-sats before returning to the central vector. The flanking probes preserve velocity conservation, traveling parallel miles out to scan the lateral void for perpendicular, blind-side hazards.'

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE SENSOR: mmWave radar (millimetre-wave) out-sees optical systems in dust and particulate — the long wavelength ignores grains that scatter light blind. That is why the phantoms fly where cameras would be useless: dense dust, debris haze, the dark between stars. THE VET'S CONSTRAINT, AFFIRMED AND SEATED: a nebula is not simply 'space dust' — density, ionisation, plasma properties and composition vary enormously, and radar propagation is determined by wavelength, particle size and concentration, and plasma/electromagnetic behaviour. So the retained claim is mmWave forward reconnaissance in particulate environments — strong, real, and honest — not guaranteed penetration of every dense nebula. The doctrine already knows: where the ionized clouds turn volatile and conductive, the ship swaps its posture (§4).

THE CYCLE: fire -> scan -> flip 180° -> retrorocket brake -> reverse -> magnetic capture is physically possible in principle — the vet graded it so — with one honest condition: the probe needs enough delta-v to execute the recovery manoeuvre. That is a propellant/trajectory calculation, not a physics contradiction, and it sits in §9 as an owed figure. THE SISTER SHIPS: the flankers preserve velocity conservation — they are ejected during a deliberate 60-minute port/starboard divergence and travel parallel miles out, so the lateral sweep costs the ship nothing in course-keeping; the mini-sats become parallel travelers scanning the perpendicular void the forward phantoms cannot see.

§3 — THE SYSTEM ON THE SHIP

- THE BOW RAIL: the forward hull tracks are the launch spine — rockets leave from the bow, so the scan always leads the ship. Cadence: one fire every 60 minutes, standing.
- THE TELEMETRY: ultra-dense, back to the helm in flight — the corridor chart grows ahead of the hull continuously, not in snapshots.
- THE RECOVERY: flip 180°, retrorocket brake, reverse into the underside pass-through open vacuum hangar bay; magnetic capture; refuel; cycle back to the bow. The bay is pass-through — the rocket

does not dock, it flows through the loop.

- THE SISTER SHIPS: every 60 minutes the ship itself diverges to port and starboard, ejecting radar mini-sats before returning to the central vector — the lateral void gets swept on the same cadence as the forward scan.

INTEGRATION NOTE (audit, flagged): the bow rail, bay and capture gear ride the hull — the 94mm stack (GP-IM-013 v1.4) stands unchanged; the rail and bay apertures are hull furniture, seated in the Phase 6 architecture. Placement and aperture specs ride the bench (§9).

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY FIGHTS WITH THESE ROCKETS)

- SCOUT VERIFICATION (the nebula doctrine): normal navigation relies on Phase 15 bow-rail radar rockets flying 1 hour ahead of the hull to flag and avoid volatile ionized gas anomalies — the phantoms are the tripwire that tells the armor-over-stealth swap when to happen.
- CROSS-AXIS VECTORING: prior to primary course acceleration, the bow rail launchers fire massive fleets of Phase 15 mmWave radar probes out of the non-flight axes — the scan is not only forward; the whole sphere gets swept before the ship commits.
- THE EARLY-WARNING PERIMETER: Phase 15/17 scout arrays establish a distributed sensor perimeter — 15-minute to 1-hour tactical warning of storm-density plasma and charged-particle fronts.
- SECOND LIFE: captured orbital scrap is fitted with Phase 15 directional actuators and steered into pinpoint landings on un-inhabited, zero-atmosphere moons — the phantom drive unit is a standard part beyond the rocket itself.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the rocket around the radar: the mmWave array is the payload and the purpose; airframe, tank and retrorocket stage wrap around it, not the reverse.
- STEP 2 — Fit the telemetry plant: ultra-dense real-time downlink to the helm — the rocket is a sensor with an engine, and its product is data, not hardware preservation. Reuse is the economy; the data is the mission.
- STEP 3 — Install the bow rail tracks on the forward hull; commission the fire cadence controller (60 minutes, standing) with manual helm override.
- STEP 4 — Build the underside pass-through bay: open vacuum hangar, through-flow, magnetic

capture cradle, refuel coupling, and the cycle trunk back to the bow. Test the capture loop with dead-weight rounds before live propellant.

- STEP 5 — Commission the sister-ship divergence: the 60-minute port/starboard maneuver, the mini-sat ejectors, and the velocity-conservation bookkeeping at the helm so the lateral sweep never costs the course.
- STEP 6 — Bench the recovery delta-v budget end-to-end (§9) and seat the figures; only then call the loop operational.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 14 — Piezoelectric Acoustic Phase-Inversion Defensive Shields (GP-IM-014 v1.3): the cluster mate — 14 answers what arrives as vibration; 15 sees it coming first.
- PHASES 16-18 — the cluster mates: 16's station-kept data-highway buoys carry the phantom telemetry's long haul; 17's debris beacons join 15 in the early-warning perimeter (15/17); 18's radial array joins 15 as the detection pair (15/18).
- PHASE 48 — the nebula doctrine (the armor-over stealth swap): the phantoms fly one hour ahead as the tripwire for the volatile ionized clouds.
- PHASE 6 — Hull & Structure (GP-IM-006 v1.4): the bow rail, the pass-through bay and the capture loop seat in the hull architecture.
- PHASE 13 — the 94mm stack (GP-IM-013 v1.4): the hull the rail and bay ride on; unchanged by this phase.

§8 — DO-NOT-BUILD

- NEVER sell mmWave as 'through every dense nebula' — the vet constraint is seated law of reading: particulate environments yes; every nebula, no (§2).
- NEVER treat the recovery loop as free — no operational cadence until the delta-v budget is benched and seated (§9).
- NEVER let the sister-ship divergence drift the course — velocity conservation is the design's own law; the helm books it every cycle.
- NO optical-only substitution in dust — the mmWave array is the point of the phase; cameras ride along, they do not replace.

§9 — OWED

THE VET'S OPEN LIST, STILL OPEN (seated verbatim): probe mass, radar power, range, Δv and recovery trajectory. Plus, this chapter's bench: bow-rail and bay placement and aperture specs on the 94mm hull (§3); sister-sat unit count and endurance; the capture-loop's dead-weight test figures. All owed items ride the bench until spoken.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-015, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 20 August 2026.

PROVENANCE — HIS ORDER, VERBATIM: 'add the silicone to stores and tools as well then 15'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Next version triggers: the owed figures of §9, rail/bay placement rulings, or his word.

END OF CHAPTER 15